PRACTICAL CONCEPT / ABOUT 10 MINUTES

2026-10-19 | VS&A Daily |

The damping your window added

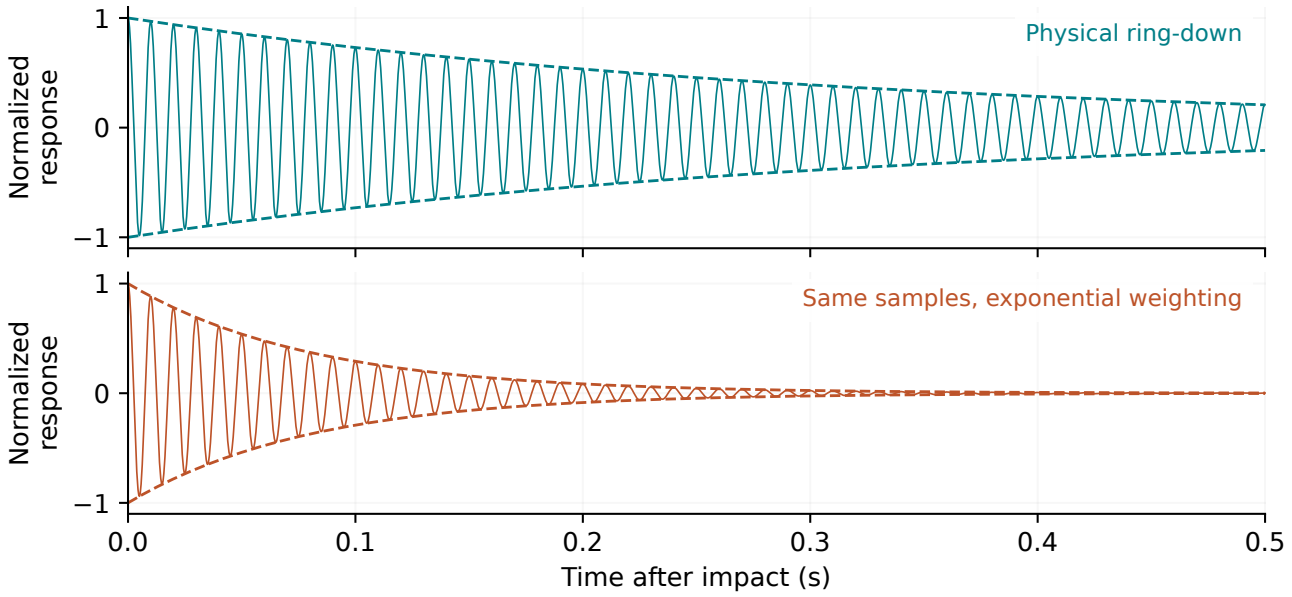


Figure 1. Original illustrative signals, not measured hardware data. Both traces begin with the same 100 Hz, 0.50%-damped mode. Only the processing changes. Dashed lines show decay envelopes; the window reaches 1% at 0.5 s.

The joint has not changed. Neither has the accelerometer, suspension, or impact point. Yet the damping estimate has nearly quadrupled after one processing setting was changed. The frequency still looks right, so the result can pass a quick plausibility check.

This is the trap of an exponential response window. It helps a finite record look like a completed transient by forcing its tail downward. That is useful, but the faster decay is partly something the analyst supplied. A smoother frequency-response function is not necessarily a more faithful picture of the physical damping.

Today’s objective is to quantify that added decay, remove it correctly when the assumptions permit, and recognize what correction cannot recover. The lesson builds on pre-test planning by following the data one step farther: from an apparently clean impact record to a damping value that may enter a response model or design review.

The central distinction is simple: the exponential window adds a *decay rate*, with units of inverse seconds. It does not add the same damping-ratio percentage to every mode. Keeping those two quantities separate prevents the most consequential bookkeeping mistake.

01 / A FINITE RECORD IS NOT AN INFINITE RING-DOWN

Why the window is tempting

A lightly damped structure may still be ringing when the acquisition block ends. The DFT represents that finite block through a periodic extension. If the end does not join the beginning cleanly, the resulting spectral leakage spreads information across frequency lines. An exponential window reduces the remaining tail before the record is transformed [1,2].

Let the post-impact motion of an isolated underdamped mode have amplitude A, phase φ, decay rate σ, and damped angular frequency ω_d:

$$x(t) = Ae^{-\sigma t}\cos(\omega_d t + \phi)$$

$$\sigma = \zeta\omega_n \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}$$

Here ω_n is the undamped natural angular frequency in rad/s, and ζ is the dimensionless damping ratio. Choose a response window that starts at unity and has end fraction r after duration T:

$$w(t) = e^{-\alpha t} \qquad \alpha = -\frac{\ln r}{T}$$

For $0 < r \le 1$, the window decay rate α is nonnegative and has units s^{-1}. Multiplication adds the exponents, so the apparent decay rate becomes $\sigma + \alpha$, while ω_d remains unchanged. In pole language, the complex-conjugate poles move left by α in the s-plane.

That does not mean every reported “frequency” is exactly unchanged. A fitted undamped natural frequency depends on both decay rate and damped frequency. A spectral-peak location is yet another quantity, influenced by the response type and neighboring modes. For light damping these differences can be tiny, while the damping error is large.

The mathematics is not new. Richardson’s modal-analysis seminar paper described exponential weighting as adding known modal decay and warned that it can merge closely spaced resonances [1]. The useful lesson is not to ban the window, but to preserve its definition and account for what it does.

02 / THE SAME MODE, PROCESSED TWO WAYS

Half a percent becomes nearly two

Take an illustrative linear mode with natural frequency 100 Hz and damping ratio 0.005. Its physical decay rate is 3.1416 s^{-1}. Record 0.5 s of motion and apply a window whose end fraction is 0.01. The window contributes 9.2103 s^{-1}, so the processed decay rate is 12.3519 s^{-1}.

Quantity	Physical mode	After weighting
Damped frequency	99.99875 Hz	99.99875 Hz
Decay rate	3.1416 s^{-1}	12.3519 s^{-1}
Fitted natural frequency	100.0000 Hz	100.0181 Hz
Fitted damping ratio	0.5000%	1.9655%
Envelope at 0.5 s	20.79%	0.2079%

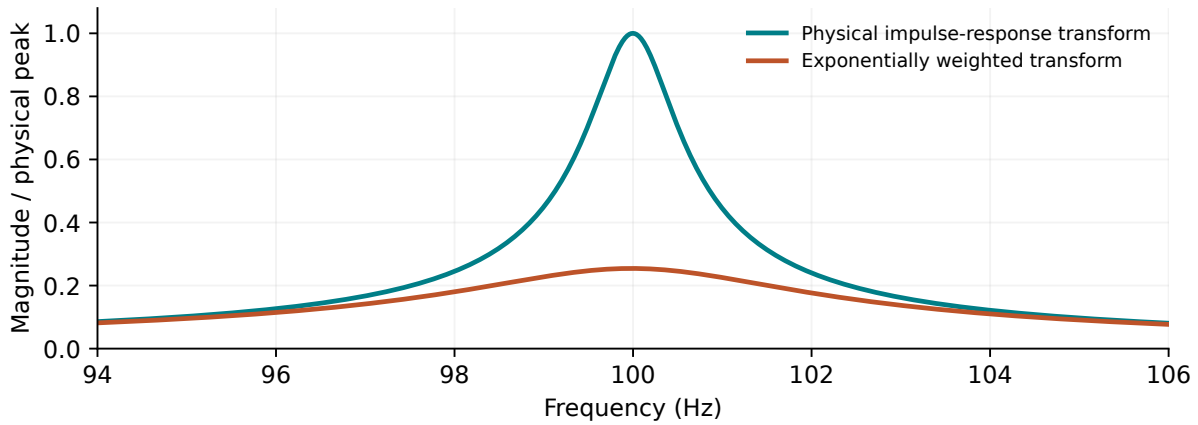


Figure 2 and table. Original analytic calculation. Curves are continuous transforms of an ideal SDOF displacement impulse response, before and after exponential weighting, normalized to the physical peak. They isolate the window effect; they are not finite-record FFT estimates or measured FRFs.

The window makes the tail more than comfortable for display, but the broadened peak now describes the weighted response. If the uncorrected 1.97% were transferred into a physical FEM in place of 0.50%, a lightly damped, single-mode resonant response under the same harmonic modal force could be underpredicted by roughly a factor of 3.9. That consequence follows from the inverse-damping scaling near resonance, not from any measured flight result.

The frequency agreement is therefore weak reassurance. A correlation table can look excellent in hertz while carrying a seriously misleading damping assumption into the next analysis.

03 / CORRECT THE POLE, NOT THE PICTURE

Subtract in the right units

Suppose a fit to the weighted response returns the positive decay rate σ_w and damped angular frequency ω_d. For the ideal exponential weighting used here, recover the physical pole before forming its damping ratio:

$$\sigma = \sigma_w - \alpha$$

$$\omega_n = \sqrt{\sigma^2 + \omega_d^2} \qquad \zeta = \frac{\sigma}{\sqrt{\sigma^2 + \omega_d^2}}$$

In the example, subtracting 9.2103 from 12.3519 s⁻¹ recovers 3.1416 s⁻¹ and hence 0.50% damping. For light damping, the familiar approximation subtracts α/ω_n from the apparent damping ratio. The pole-based form makes the assumptions and units clearer, especially when the added decay is not small.

A numerical correction is not an uncertainty correction. If the fitted decay rate has an uncertainty of ±0.6 s⁻¹, subtracting a precisely known window rate leaves approximately the same absolute uncertainty in the physical decay rate. Around this example's 3.14 s⁻¹, that is about 19% relative uncertainty. Subtracting two relatively large numbers can expose how little physical damping was actually resolved.

Nor does correction undo the loss of information caused by suppressing late-time data. It cannot restore clipped samples, separate modes that the measurement cannot resolve, or turn a nonlinear decay into a single constant damping ratio. If the corrected rate is negative for a supposedly passive, stable test article, investigate the fit, window convention, noise and test conditions before treating the result as a physical instability.

For a true LTI impulse response, exponential weighting corresponds to evaluating its Laplace transform at $s + \alpha$. For a finite-duration impact, the measured force and response must be processed consistently; a response-only window is not automatically the same operation on the FRF. Fladung and Rost explicitly examined the input-window issue [2]. The table deliberately assumes an ideal impulse to isolate the pole shift.

Check whether your software already compensates. Siemens documents automatic exponential-window correction in its modal fitting workflow and certain damping displays [3]. That is a product-specific behavior, not permission to assume every exported curve or third-party fit is corrected. Applying the correction twice is another error.

04 / BUILD A DAMPING MEASUREMENT YOU CAN DEFEND

Change the record before changing the structure

Start with unweighted time histories. Confirm a clean single impact, adequate force bandwidth, no clipping, and a response above the noise floor. Examine the actual tail, not merely the plot’s autoscaled appearance. Trigger location and pretrigger duration matter because they determine where the window begins relative to the event.

Where practical, lengthen the acquisition so the physical response decays sufficiently without aggressive weighting. Longer records also reduce the native FFT line spacing. For the example’s physical decay rate, reaching a 1% envelope requires about 1.47 s. That is a calculated duration for this ideal mode, not a universal record-length rule. Additional recording is useful only while the response remains informative relative to noise and other disturbances.

Preserve raw force and response data, sample rate, record duration, trigger timing, exact force and response windows, and the correction status of exported modal parameters. “Exponential: 1%” is incomplete without its time basis and software convention. A finite exponential never reaches exactly zero; a displayed “0%” setting needs its implementation definition.

Then perform a controlled sensitivity check. Fit the same raw records using a longer usable block and several defensible window strengths, keeping the modal model and fit band traceable. Corrected physical poles should be reasonably consistent within the measurement uncertainty. A damping estimate that tracks the window setting is a warning, not evidence that the joint dissipates more energy.

Avoid solving a resolution problem by zero-padding. Padding provides denser samples of the same finite-record spectrum; it adds no measured ring-down. If close modes dominate the band, use a multi-mode fit and enough independent response information to identify them. Do not substitute a single half-power bandwidth for an unresolved modal cluster [1,3].

ISO 7626-5:2019 covers impact-excited mobility and related FRFs, including driving-point and transfer measurements, using an unattached exciter. Its scope is not a general acceptance standard for every modal survey, and it does not make this lesson’s numerical settings universal requirements [4]. Define the needed damping accuracy from the downstream decision: a preliminary frequency survey and a resonance-sensitive load prediction need different evidence.

The review question is worth asking plainly: how much of the reported decay came from the structure, and how much came from the processing? A traceable answer is more valuable than a visually tidy peak.

05 / CARRY THE METADATA WITH THE MODES

Three takeaways

1. A window can imitate damping. Exponential weighting adds a decay rate, while leaving the damped oscillation frequency unchanged in the ideal modal signal.

2. Correct the pole once. Subtract the known rate in s^{-1}, then calculate the physical damping ratio. Verify whether the software has already done it.

3. Correction cannot recreate evidence. Preserve raw records and window metadata, assess uncertainty, and test sensitivity to duration and weighting before using damping in prediction.

A short reflection

Two modes at 50 and 200 Hz use the same exponential window. Does the window add the same damping-ratio increment to both?

Answer: No. It adds the same decay rate. For light damping, the ratio increment is approximately the window rate divided by modal angular frequency, so the 50 Hz mode receives roughly four times the increment of the 200 Hz mode. This comparison assumes both are well identified and the same weighting is applied.

Sources and figure notes

[1] M. Richardson, “Modal Analysis Using Digital Test Systems,” Seminar on Understanding Digital Control and Analysis in Vibration Test Systems, Part 2, 1975 seminar, pp. 43-64; exponential smoothing on pp. 53-54 (PDF pp. 56-57). NASA-hosted primary proceedings.

[2] W. Fladung and R. Rost, “Application and Correction of the Exponential Window for Frequency Response Functions,” Mechanical Systems and Signal Processing 11(1), 1997, pp. 23-36. Laplace-shift correction and force/response window consistency.

[3] Siemens, How to Calculate Damping from a FRF?, sections 5.1-5.3. First-party software guidance on resolution, exponential-window correction, and mode spacing. Undated page; accessed October 19, 2026.

[4] ISO 7626-5:2019, Mechanical vibration and shock - Experimental determination of mechanical mobility - Part 5, edition 2, December 2019. Public scope consulted; no inaccessible clauses are asserted.

Figures 1-2 and the table are original calculations, not measured data. The opening visual is an illustrative signal comparison, not a historical photograph. Model preference: this run could not verify or select GPT-6 Astra.